NFL Defensive Coverage Predictor

CS/SE 4AL3 – Project Proposal

Group 77

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Task Overview and Significance

Our task is to recognize defensive coverages before the snap. This is challenging because defenses often try to disguise their coverage by showing one formation or alignment before the snap and executing another after the snap. This makes it difficult for an offensive player or coordinator to determine the defense's true coverage based only on its pre-snap alignment.

This matters because accurately identifying defensive coverage before the play begins could help offensive coordinators and quarterbacks make better pre-snap decisions, such as adjusting the play, route combinations, or blocking scheme.

This project is non-trivial because the model must make its prediction using only pre-snap information, while accounting for defensive disguises, player positioning, and different formations. We will investigate which pre-snap features are most useful for distinguishing between different defensive coverages.

Task Definition

Given the pre-snap alignment of both teams and the game situation, the model outputs the coverage scheme the defense will play.

- **Input/data type:** Tabular. Each sample is the last complete tracking frame before `ball_snap`, normalized so the line of scrimmage is $x = 0$ and the offense moves left to right: (x, y), speed, acceleration,

and orientation of all 22 players, plus down, distance, yardline, clock, score differential, and offensive formation.

- **Prediction task:** Supervised multi-class classification.
- **Labels and output:** One categorical label per play, the defense's coverage from the 12 observed PFF `pff_passCoverage` classes (Cover-0, Cover-1, Cover-2, 2-Man, Cover-3, Quarters, Cover-6, Bracket, Goal Line, Red Zone, Prevent, and a catch-all Miscellaneous). Very sparse classes may be merged into parent schemes, or we may evaluate a coarser Man vs. Zone schema.
- **Unit of prediction:** One passing play. No post-snap frames, route data, `pff_passCoverageType`, or other post-snap annotations are given to the models, to prevent target leakage.

Data Sources and Collection Plan

We downloaded the Big Data Bowl regional-event dataset: 122 games from weeks 1–8 of the 2021 season. Its documentation identifies NFL Next Gen Stats as the tracking provider and Pro Football Focus (PFF) as the scouting and coverage-label provider. Next Gen Stats collects player movement through stadium receivers and RFID tags at 10 samples per second. The NFL organizes the release; detailed PFF annotation procedures are not documented.

We obtained CSVs directly from GitHub after the original Kaggle 2025 dataset was removed. No scraping or API polling is needed. Raw data stays in the ignored `data/regional-2021/` directory; a manifest records the source revision and file hashes. Data-use and redistribution terms remain to be verified.

The release contains game, play, player, PFF scouting, and per-game tracking tables. We join plays and tracking on `gameId`/`playId`, and player metadata on `nflId`. Available metadata includes week, teams, player positions, timestamps, direction, and snap events. Inputs will combine player coordinates, speed and acceleration with down, yards to go, clock, and pre-snap scores. The existing `pff_passCoverage` labels provide 12 observed classes; no new annotation is required.

We will retain the last complete frame strictly before each play's `ball_snap`, with 11 distinct players per team and valid coordinates, then normalize direction and positions relative to the line of scrimmage. One snapshot per play keeps the input small while preserving alignment. We exclude coverage annotations, post-snap frames, descriptions, and outcomes from predictors. Games stay together across splits; preprocessing is fitted on training data only.

Dataset size: There are 8,557 plays and 8,314,178 tracking rows. The initial audit retained 8,522 labeled

plays with complete pre-snap snapshots and excluded 35 that failed these checks. Plays, rather than frames, count as instances; further quality checks may reduce this total.

Three actual labeled examples: Inputs contain all 22 player coordinates and game context. Below are compact summaries of three source records: down/distance, defensive coordinate ranges in yards, and the recorded PFF label.

1. Game 2021090900, play 97, frame 5: down 3, 2 yards to go; defensive x range 43.38–58.88, y range 15.98–37.61 yards. **Label:** Cover-1.
2. Game 2021090900, play 137, frame 6: down 1, 10 yards to go; defensive x range 91.82–108.55, y range 8.77–43.75 yards. **Label:** Cover-3.
3. Game 2021090900, play 456, frame 5: down 2, 10 yards to go; defensive x range 31.23–45.91, y range 10.47–44.64 yards. **Label:** Cover-6.

Proposed Solution and Evaluation

Features. Engineered from the pre-snap snapshot rather than raw coordinates:

- **Deep shell:** deepest safety depth, lateral split of deep safeties, single-high vs. two-high.
- **Box count:** defenders within 5 yards of the line of scrimmage and inside the offensive tackles.
- **Cornerback cushion:** distance between outside cornerbacks and the widest receivers.
- **Receiver matchups:** nearest-defender distance and angle for each eligible receiver.

Models.

1. Majority-class baseline (always predict the most frequent training class).
2. Primary models: Random Forest and gradient-boosted trees (XGBoost/LightGBM), which capture threshold interactions such as safety split with cornerback cushion.

Existing solutions and how we differ.

- **Michels et al.:** Man vs. Zone using pre-snap motion and Hidden Markov Models. We predict the 12-class PFF schema from static alignment.
- **NFL Next Gen Stats/AWS (H. Song et al., 2023):** 8 coverages from the whole play. We use only pre-snap information and report interpretable feature importance.
- **Dutta et al.:** unsupervised clustering of defensive backs. We use supervised learning on PFF labels.

Planned libraries: pandas, numpy, scikit-learn, xgboost, lightgbm, shap, matplotlib, seaborn.

Evaluation:

- **Splits:** by game ID (about 70/15/15 train/validation/test), so no game appears in more than one split. Preprocessing is fitted on training data only.
- **Metrics:** macro-averaged F1 and per-class recall (so common coverages do not dominate), plus log-loss, accuracy, a normalized confusion matrix, and Man vs. Zone accuracy for comparison with the literature.
- **Success criteria:** macro-F1 clearly above the majority baseline, and Cover-1 vs. Cover-3 and Cover-2 vs. Quarters separated better than chance.

Sources inspiring the approach

1. R. Michels, R. Bajons, J.-O. Fischer. *Integrating Unsupervised and Supervised Learning for the Prediction of Defensive Schemes in American football*. arXiv:2602.10784, 2026.
2. H. Song et al. *Identifying defense coverage schemes in NFL's Next Gen Stats*. AWS Machine Learning Blog, Feb. 2023.
3. R. Dutta, R. Yurko, V. Ventura. *Unsupervised Methods for Identifying Pass Coverage Among Defensive Backs with NFL Player Tracking Data*. arXiv:1906.11373, 2019 (rev. 2020).

TA Consultation

We consulted with Abdelaziz Serour twice throughout the development of our project idea. During our first interaction, our team proposed developing a SpecDec system for tool calls. However, after discussing the idea with the TA, we concluded that it was too complex and theoretical for the scope of the project. As a result, we decided to pivot toward a more practical and applied idea.

We then developed the idea of recognizing defensive coverages during the pre-snap period. We reached out to Abdelaziz again through Outlook to discuss whether this idea was feasible, and he agreed that it was a suitable project that we could proceed with.